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Reply to 'Musical neurodynamics and the 'inner voice''

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We thank Stone and Buks for their interest in our Perspective (Harding, E. E. et al. Musical neurodynamics. *Nat. Rev. Neurosci.* **26**, 293–307; 2025)[1], and for the opportunity to clarify several points of interpretation in their Correspondence (Stone, L. & Buks, E. Musical neurodynamics and the 'inner voice'. *Nat. Rev. Neurosci.* https://doi.org/10.1038/s41583-026-01041-5; 2026)[2].

Stone and Buks suggest that our discussion implies that locally stable states in nonlinear dynamical systems should spontaneously 'gravitate' towards simpler integer ratios[2]. This interpretation does not reflect our intent. As is well established in dynamical systems theory, multiple locally stable resonances can coexist, including higher-order ratios such as 3:2, which – once established – do not transition to other states in autonomous systems (that is, systems whose governing equations do not depend on time or on external input)[3,4]. Neural resonance theory (NRT) does not claim otherwise[1].

The crucial point is that neural systems engaged with music are not autonomous. They are continuously driven by structured sensory input, shaped by adaptation and plasticity and influenced by delayed feedback. In such systems, stability is not merely a binary property but reflects differences in robustness to perturbation, attractor basin geometry and resonance amplitude. In coordination dynamics, these distinctions are well documented: higher-order frequency ratios typically occupy narrower resonance regions and are therefore more easily disrupted than lower-order ratios, whereas simpler ratios exhibit broader regions of stability[5]. It is in this empirically grounded sense that we refer to some resonances as 'more stable' than others[1].

Importantly, the neural system undergoes continual perturbation as musical structure unfolds. Syncopation, expressive timing, harmonic tension and interaction among performers repeatedly displace neural dynamics away from locally stable states. Following such perturbations, the system may return to a previous resonance or reorganize around a different resonant state that is more robust under ongoing stimulation. This behaviour is naturally described in terms of structural stability; that is, the persistence of qualitative dynamical organization under parameter change[6]. In musical contexts, changes in harmonic function, key or metre can be understood as parameter modulations that reshape the attractor landscape while preserving coherent perceptual organization. For example, from this perspective, harmonic modulation corresponds not to instability, but to a controlled deformation of a structurally stable dynamical regime, enabling transitions between tonal centres without loss of musical coherence.

The publications cited in our Perspective identify relevant dynamical concepts and relate them to experimentally measurable phenomena in perception, attention and performance, yielding quantitative fits and testable predictions. For example, NRT does not posit an immutable aesthetic hierarchy of integer ratios[7]. Rather, we argue that pleasantness ratings reflect a more fundamental neural property – namely, the stability of resonant states – consistent with recent empirical findings on musical interval perception[8]. Importantly, such stability is not fixed: parameters such as coupling strength can be modified through attunement and Hebbian plasticity, as exposure and training increase the robustness of initially fragile resonances[4]. Related principles govern musical performance. Empirical studies of polyrhythmic coordination show that higher-order frequency ratios are more easily disrupted, whereas simpler ratios are more resistant to perturbation[5]. At the same time, practice reshapes attractor landscapes, enabling performers to sustain coordination patterns that would otherwise be relatively fragile[9]. On this account, both preference ratings and performance difficulty are grounded in the stability properties of underlying neurodynamic states.

We also welcome Stone and Buks's emphasis on auditory imagery and the 'inner voice'. From an NRT perspective, such phenomena need not be treated as conceptually separate from resonance-based mechanisms. Auditory imagery might arise from self-sustaining activity within learned networks of coupled neural oscillators, supported by feedback and memory. Neuroimaging evidence showing overlap between perception and imagery is consistent with this view, although formal dynamical models specifying the conditions under which imagery emerges remain an important direction for future work[10]. Fundamental dynamical principles may provide a coherent, mathematically grounded framework for understanding how musical structure, perception and auditory imagery emerge from neural and bodily dynamics.

Edward W. Large[1,2] ✉, **Eleanor E. Harding**[3,4], **Ji Chul Kim**[1], **Alexander P. Demos**[5], **Iran R. Roman**[6], **Parker Tichko**[1] & **Caroline Palmer**[7]

[1]Department of Psychological Sciences, University of Connecticut, Storrs, CT, USA. [2]Department of Physics, University of Connecticut, Storrs, CT, USA. [3]Center for Language and Cognition, University of Groningen, Groningen, The Netherlands. [4]Department of Otorhinolaryngology/Head and Neck Surgery, University Medical Center Groningen, University of Groningen, Groningen, The Netherlands. [5]Department of Psychology, University of Illinois Chicago, Chicago, IL, USA. [6]School of Electronic Engineering and Computer Science, Queen Mary University of London, London, UK. [7]Department of Psychology, McGill University, Montreal, Quebec, Canada.

✉e-mail: edward.large@uconn.edu

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Competing interests

E.W.L. is founder of, and owns stock in, Oscilloscape, Inc. (dba Oscillo Biosciences). J.C.K. is currently a paid employee of, and owns stock in, Oscilloscape, Inc. E.W.L. and J.C.K. are authors of patents owned by Oscilloscape, Inc. The subject matter of the current paper is not directly related to the business interests of Oscilloscape, and no products of Oscilloscape are discussed in this paper. The other authors declare no competing interests.